

Exercise 5 — Model-Agnostic Methods

Table of Contents

01 Introduction	3
02 Exercise 1 — 1D PDP: Bike Rental Prediction	4
Data Preparation & Model Training	4
PDP Results & Analysis	5
03 Exercise 2 — 2D PDP: Humidity × Temperature	6
Approach & Methodology	6
Interpretation	7
04 Exercise 3 — PDP: House Price Prediction	8
Data Preparation & Model Training	8
PDP Results & Analysis	9
05 Conclusions	10

Introduction

This report applies **Partial Dependence Plots (PDP)** to two real-world prediction tasks. PDP is a global model-agnostic interpretability method that visualises the marginal effect of one or two features on the predicted outcome of a previously fitted model. All models used are **Random Forests** fitted using the `randomForest` package in R.

The partial dependence function is estimated by replacing the values of the feature of interest across a regular grid while marginalising over the joint distribution of the remaining features:

PDP Formula: For each grid value x_S , the partial dependence is computed as the average prediction over all n training observations, substituting x_S while keeping all other features at their observed values. A minimum of **50 evenly spaced grid points** is used for every plot.

Rug marks displayed along the axes of each plot represent the empirical distribution of training observations, helping distinguish regions well-supported by data from potential extrapolations beyond the observed range.

EXERCISE 1

1D PDP

Bike rentals — 4 features

EXERCISE 2

2D PDP

Humidity × Temperature interaction

EXERCISE 3

1D PDP

House prices — 4 features

Exercise 1 — 1D PDP: Bike Rental

Prediction

Data Preparation and Model Training

The Capital Bikeshare dataset (day.csv) contains 731 daily aggregated rental records with weather and calendar features. A **Random Forest model** is trained on all available observations using 12 predictor variables. The variable `days_since_2011` is derived from the date column to capture temporal growth trends.

TRAINING
OBSERVATIONS

731

Full dataset used

NUMBER OF TREES

500

`ntree = 500`

OOB R^2

0.892

Out-of-bag score

PDP GRID POINTS

50

Per feature

Reproducibility note: A fixed seed (`set.seed(42)`) is applied only to the Random Forest training step, ensuring that the model produces identical results on every run. No seed is set for sampling operations, which remain truly random.

PDP Results

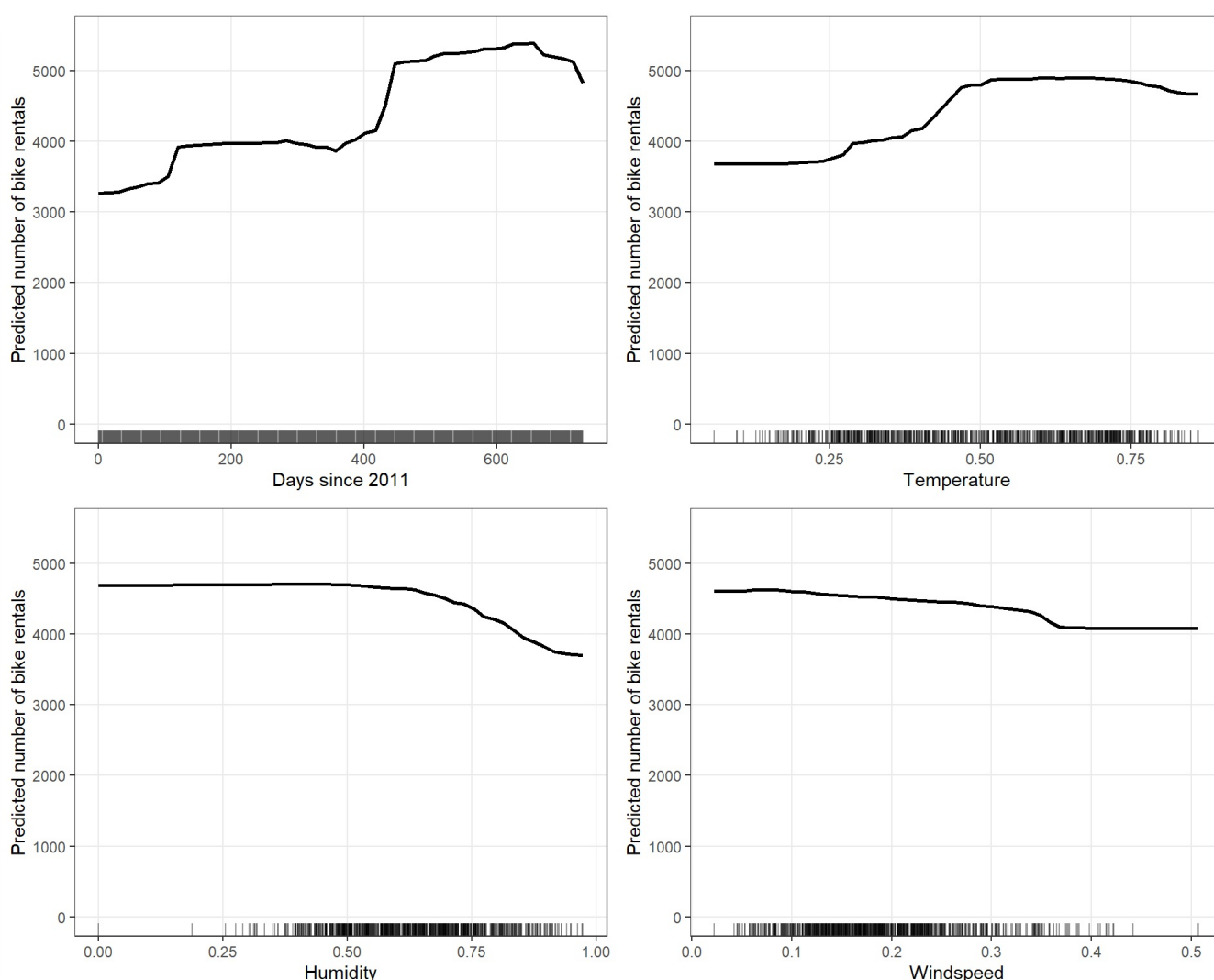


Figure 1. 1D Partial Dependence Plots for Days since 2011, Temperature, Humidity and Windspeed. Rug marks at the bottom of each panel

Analysis

- 1 **Days since 2011** — Shows a clear stepped pattern: rentals start at ~3,200, stabilise around 4,000 until day 350, then jump sharply to ~5,000–5,300 in the second year. The model captures the year-on-year growth of the service between 2011 and 2012.
- 2 **Temperature** — Presents a saturating curve: rentals grow from ~3,600 at low temperatures up to a plateau of ~4,800–5,000 at moderate-to-high temperatures (0.50–0.75), levelling off beyond that point. Mild temperatures encourage use, but the additional benefit fades beyond a comfortable cycling range.
- 3 **Humidity** — Has a threshold-based negative effect: the curve remains flat at ~4,700 rentals up to 50% humidity, after which it falls progressively to ~3,700. Humidity only penalises demand significantly once it exceeds that midpoint.
- 4 **Windspeed** — Shows a smooth and uniform negative relationship: rentals drop from ~4,650 to ~4,100 across the observed range. The impact is moderate with no clear thresholds, making it the least influential of the four variables.

Exercise 2 — 2D PDP: Humidity × Temperature

Approach & Methodology

The Random Forest model trained in Exercise 1 on all 731 rows is reused. For the 2D PDP computation, a random sample of **50 observations** is used to approximate the marginal expectation across a **50 × 50 evaluation grid** (2,500 combinations). For each cell, humidity and temperature are substituted simultaneously while predictions are averaged over the sampled rows.

GRID DIMENSIONS

50×50

2,500 evaluations

MARGINALISATION SAMPLE

50

Random rows, no seed

VISUALISATION

geom_tile

Width & height ×1.05

2D PDP - Humidity x Temperature

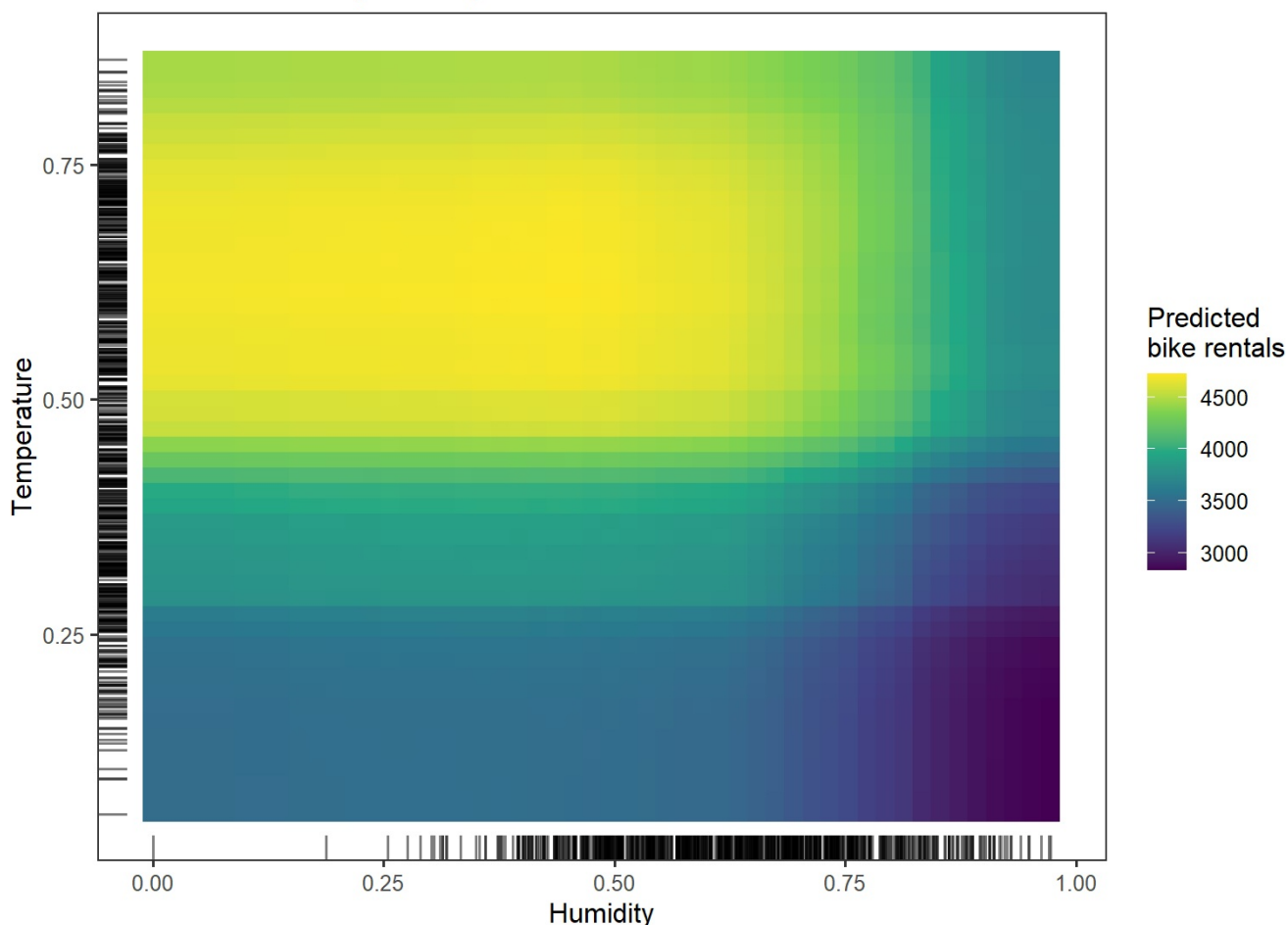


Figure 2. 2D Partial Dependence Plot for Humidity and Temperature. Yellow regions indicate high predicted demand (~4,500 rentals); dark purple regions indicate low demand (~3,000 rentals). Rug marks on both axes show the empirical distribution of training observations.

Interpretation

The heatmap gradient is predominantly **horizontal**, indicating that **temperature dominates over humidity** in driving predicted bike demand. The yellow regions cover the entire band of high temperatures (>0.50) regardless of humidity level, while the dark regions correspond uniformly to low temperatures.

Humidity plays a **secondary role**: its effect is only visible at high temperatures, where values above 0.75 slightly

reduce predicted demand. Below that threshold, humidity has negligible additional impact.

The rug marks reveal that real observations concentrate at medium-to-high humidity and moderate temperatures. The extreme corners of the plot — very dry and very warm, or very wet and very cold — are less supported by actual training data and should be interpreted with caution.

Exercise 3 — PDP: House Price Prediction

Data Preparation and Model Training

The King County house price dataset (kc_house_data.csv) contains 21,613 property sale records. Due to its size, a **single random subset of 2,000 rows** is drawn from the full dataset. This same subset is used both to train the Random Forest and to compute the PDPs, keeping the approach consistent and computationally tractable.

FULL DATASET	SUBSET USED	OOB R^2	FEATURES TRAINED
21,613	2,000	0.609	6
Total rows available	Train & PDP (no seed)	Out-of-bag score	bedrooms, bathrooms, sqft, lot, floors, yr_built

PDP Results

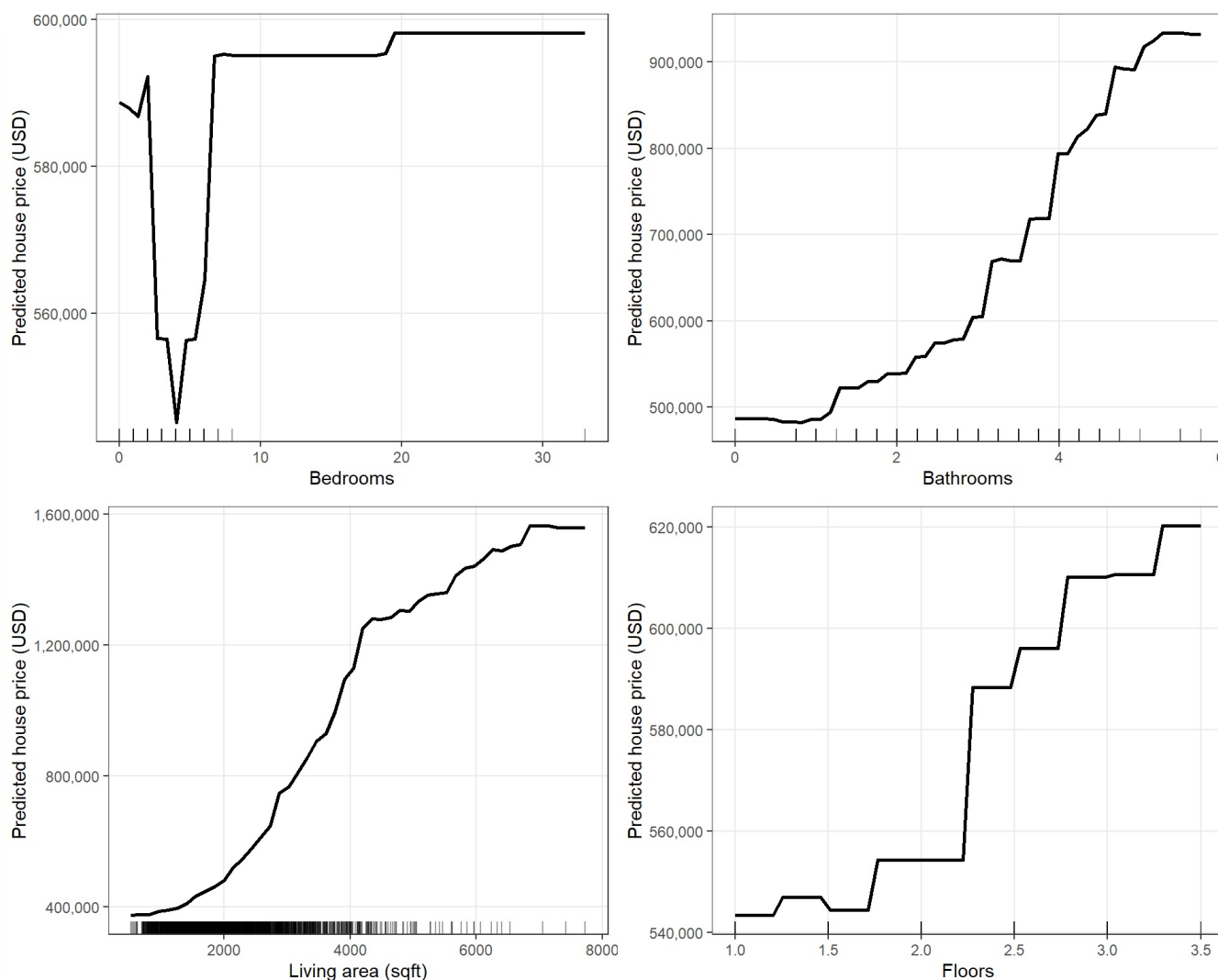


Figure 3. 1D Partial Dependence Plots for Bedrooms, Bathrooms, Living Area (sqft) and Floors. Rug marks at the bottom of each panel show the distribution of the 2,000 training observations.

Analysis

- 1 Bedrooms** — Shows anomalous behaviour: prices start at ~\$590,000, drop sharply to a minimum around 3-4 bedrooms (~\$548,000), then rise and stabilise at ~\$595,000 from 7-8 bedrooms. The initial drop is likely a model artefact caused by data scarcity at extreme bedroom counts, as confirmed by the rug marks

concentrating between 1 and 6 bedrooms.

- 2 **Bathrooms** — Presents a clear positive stepped relationship: predicted price rises from ~\$490,000 with few bathrooms to ~\$930,000 with 5–6 bathrooms. The staircase shape reflects the discrete nature of bathroom counts. With a total range of ~\$440,000, it shows the largest absolute price impact of the four features.
- 3 **Living Area (sqft)** — Shows the clearest relationship: monotonic growth from ~\$370,000 for small homes to ~\$1,600,000 for the largest. Growth is steepest between 1,000 and 4,000 sqft where most observations concentrate, gradually levelling off for very large properties.
- 4 **Floors** — Produces a stepped pattern reflecting its few discrete values (1, 1.5, 2, 2.5, 3, 3.5). Prices rise from ~\$543,000 for single-storey homes to ~\$620,000 for 3.5-storey ones, with the largest jump between 1 and 2 floors. The effect is positive but modest.

Conclusions

KEY FINDINGS

Bike rentals (1D PDP): Time shows a stepped year-on-year effect reflecting service growth. Temperature acts as a saturating curve. Humidity only penalises demand above 50%. Windspeed has a moderate uniform negative impact.

Bike rentals (2D PDP): Temperature clearly dominates over humidity. The colour gradient is primarily driven by temperature level; humidity plays a secondary role visible only at high temperatures.

House prices: Living area is the most influential variable. Bathrooms show the largest absolute price impact. Floors have a positive but modest effect. Bedrooms display anomalous behaviour linked to data scarcity at extreme values.

Key limitation — Marginal Independence Assumption: When features are correlated, the PDP grid evaluation integrates over feature combinations that may not exist in practice, potentially distorting the estimated effect. In the house price dataset, where bedrooms, bathrooms and living area are naturally correlated, **Accumulated Local Effects (ALE) plots** would offer a more robust alternative.